Ashoka University
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Braess's Paradox in Transportation Networks

Term Paper for CS-4231: Graph Theory and Network Sciences (Spring 2026)

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Code Repository:

`https://github.com/neha-palak/braess`

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Abstract

Braess's paradox, discovered in 1968, reveals a counterintuitive phenomenon in traffic flow: adding capacity to a congested network can paradoxically increase overall travel times when drivers make selfish routing decisions. This study investigates how individual optimization at the microscopic level leads to collective outcomes at the macroscopic level. Through network simulation using the Sioux Falls transportation network dataset, we demonstrate that network modifications alter the equilibrium distribution of traffic flows, potentially degrading system performance. We model driver behavior using Wardrop's equilibrium principle, wherein all routes in use have equal travel cost, implementing this as a Nash equilibrium problem. Our analysis employs the Bureau of Public Roads (BPR) latency function to capture congestion effects. We simulate selfish routing using iterative traffic assignment algorithms and compare total network travel times before and after adding a critical edge. (After simulation we can confirm:) Our findings confirm that Braess-like effects can emerge in transportation networks, illustrating the gap between individual rationality and collective optimization. Finally, we discuss the extensions of this paradox to power grids and communication networks, demonstrating its broader applicability across infrastructure systems.

Introduction

What is Braess's Paradox?

Braess's paradox is a remarkable result in network flow theory, first observed by the German mathematician Dietrich Braess in 1968. The paradox demonstrates that in certain network configurations, adding an additional edge—such as a new road in a transportation network—can decrease network efficiency and increase the average travel time for all users, even though the added capacity should intuitively improve performance. Consider a simple example: suppose two routes exist between a source and destination, both with fixed and variable latency functions. When traffic equilibrates according to user optimization (each driver chooses the shortest available route), drivers distribute themselves such that all routes in use have equal travel cost. Now suppose we add a new, very fast zero-cost shortcut between two intermediate nodes. Drivers will exploit this shortcut, but because all drivers behave this way simultaneously, the new route becomes congested. The equilibrium shifts, and paradoxically, the overall system travel time increases beyond the original state. This occurs because the system moves to a different Nash equilibrium—one that is individually rational for each driver but collectively worse for the entire population. The paradox fundamentally challenges the intuition that more infrastructure is always beneficial. It reveals a tension between individual optimization (selfish routing) and systemwide optimization (social welfare). This tension is quantified by the price of anarchy, a metric that measures the performance degradation of a system operating at Nash equilibrium compared to the social optimum (Can use the illustration we used in the presentation in this section)

Why Braess's Paradox Matters

Braess's paradox has significant implications across multiple domains:

(i) **Urban Transportation**: Traffic congestion in modern cities is a critical problem affecting billions of dollars in lost productivity and environmental degradation. Urban planners often invest in road expansion, assuming that added capacity will reduce congestion. However, Braess's paradox suggests that such investments may backfire, especially when they create attractive shortcuts. This phenomenon has been documented in real cities, most notably when Seoul removed an elevated highway in 2003, and traffic speeds actually improved—an event sometimes attributed to Braess-like effects. Understanding this paradox helps planners make more informed infrastructure decisions and consider congestion management policies (such as tolling or route guidance) instead of blind capacity expansion.

(ii) **Network Design and Optimization**: The paradox illustrates a fundamental principle in network optimization: the global optimum may require restricting certain edge capacities or even removing edges. This insight has applications in telecommunications, where routing protocols must balance load distribution with avoiding congestion collapse. Similarly, in supply chain networks, adding distribution centers can sometimes increase overall delivery times if not carefully planned.

(iii) **Game Theory and Mechanism Design**: The paradox exemplifies the broader concept that individual rationality does not guarantee collective rationality. This principle underpins research in algorithmic game theory, mechanism design, and the price of anarchy—all critical for understanding emergent behavior in decentralized systems.

(iv) **Policy and Infrastructure Planning**: Recognizing Braess's paradox motivates alternative transportation management strategies such as congestion pricing, demand management, and public transit investment—approaches that might be more effective than simple capacity expansion.

Literature Survey

Foundational Works and Historical Context

(i) Braess (1968) introduced the paradox through a simple network example, demonstrating mathematically that equilibrium travel time could increase after adding a zero-cost edge. His original paper 'On a paradox in traffic planning', established the theoretical foundation for understanding how user behavior can lead to counterintuitive outcomes.

(ii) Wardrop (1952) developed the traffic equilibrium model that underlies modern transportation analysis. Wardrop's First Principle states that traffic arranges itself so that no driver can unilaterally reduce their travel time by changing routes. This principle, now termed Wardrop equilibrium, is equivalent to the Nash equilibrium concept in game theory. Wardrop's framework provides the equilibrium condition necessary to analyze how networks respond to capacity changes.

(iii) Pigou (1920) developed early congestion cost models, distinguishing between private costs (experienced by an individual traveler) and social costs (imposed on the entire system). The famous "Pigou problem"—a simple two-route network demonstrating the gap between individual and social optima—is a foundational example in congestion game theory and directly relates to Braess's paradox.

Modern Developments and Extensions Roughgarden (2005), in Selfish Routing and the Price of Anarchy, provided a comprehensive framework for analyzing the efficiency loss of selfish routing. Roughgarden quantified the price of anarchy—the ratio of cost at Nash equilibrium to the cost at the social optimum—for various network topologies and latency function classes. His work established that for cer-

tain latency functions (such as linear functions), the price of anarchy is bounded, placing limits on how badly selfish routing can perform.

Theoretical Background

We represent a transportation network as a directed graph, where: V is the set of nodes (representing intersections or locations) E is the set of edges (representing roads or links). Each edge has an associated latency function , which specifies the travel time on edge as a function of the traffic flow using that edge. For a path from source to destination , the total latency is the sum of edge latencies: (LaTex equation for total latency as a sum of edge latencies)

An Origin-Destination (OD) pair has a fixed demand —the total volume of traffic that must flow from to . The traffic flow must satisfy flow conservation constraints and be distributed among available paths.

Wardrop's First Principle states that at equilibrium: All paths carrying positive flow have equal and minimal latency All other paths have latency greater than or equal to the minimum Formally, a feasible flow is a Wardrop equilibrium if for all OD pairs : This principle is equivalent to the concept of Nash equilibrium: no individual driver can unilaterally improve their travel time by switching to a different route, assuming all other drivers' routes remain fixed. Wardrop equilibrium is the natural outcome when each driver independently chooses the route that minimizes their personal travel time. Unlike a centrally planned optimum (which would minimize total system latency), Wardrop equilibrium reflects how traffic actually organizes itself in decentralized transportation systems. Under continuous, strictly increasing latency functions (a standard assumption), a Wardrop equilibrium always exists. However, equilibrium is not always unique; multiple equilibria may exist depending on network structure and latency functions.

Latency Functions: The BPR Function

The *Bureau of Public Roads (BPR) function* (1) is the standard latency model used in transportation engineering:

$$\ell_e(f_e) = t_0^e \left(1 + \alpha \left(\frac{f_e}{c_e}\right)^\beta\right) \tag{1}$$

where:

- t_0^e = free-flow travel time on edge e (the time to traverse the edge with zero traffic)
- f_e = current traffic flow on edge e
- c_e = capacity of edge e
- α, β = congestion parameters (typically $\alpha = 0.15$ and $\beta = 4.0$ in real networks)

Interpretation

- When $f_e = 0$, latency equals t_0^e
- As f_e approaches c_e, latency increases non-linearly
- The parameter β controls how quickly congestion increases; higher β means sharper congestion effects near capacity

Properties

The BPR function possesses the following desirable properties:

(i) **Continuous differentiability:** The function is smooth and differentiable with respect to f_e

(ii) **Strict monotonicity:** $\ell_e(f_e)$ is strictly increasing in flow, i.e.,

$$\frac{\partial \ell_e}{\partial f_e} = t_0^e \cdot \alpha \cdot \beta \left(\frac{1}{c_e}\right)^\beta f_e^{\beta-1} > 0 \quad \forall f_e > 0 \tag{2}$$

(iii) **Convexity:** The second derivative is positive, meaning marginal latency increases with flow:

$$\frac{\partial^2 \ell_e}{\partial f_e^2} = t_0^e \cdot \alpha \cdot \beta(\beta - 1) \left(\frac{1}{c_e}\right)^\beta f_e^{\beta-2} > 0 \quad \text{for } \beta > 1 \tag{3}$$

The convexity property is especially important because it reflects realistic traffic behavior: each additional vehicle causes increasing delays. This property also plays a critical role in determining when Braess's paradox can occur.

Practical Relevance

The BPR function has been calibrated using empirical transportation data across many cities and networks. The parameter values ($\alpha = 0.15, \beta = 4.0$) are widely adopted in transportation modeling practice. This makes the BPR function ideal for simulations that aim to reflect real-world traffic dynamics, as demonstrated in our analysis of the Sioux Falls network.

Social Optimum vs. Nash Equilibrium

A central problem in congestion game theory is the gap between the *Nash equilibrium* (outcome of selfish routing) and the *social optimum* (outcome minimizing total cost).

Social Optimum

The social optimum seeks to minimize the total system travel time (also called the total system cost or social cost). Formally, we seek a feasible flow f^* that minimizes:

$$\mathrm{SC}(f^*) = \min_f \sum_{e \in E} f_e \cdot \ell_e(f_e) \tag{4}$$

subject to flow conservation constraints:

$$\sum_{P: e \in P} f_P = d_{st} \quad \forall \text{ OD pairs } (s,t) \tag{5}$$

The social optimum represents the outcome if a central authority could assign all drivers to routes in a coordinated manner. At the social optimum, the marginal cost of using an edge (including congestion externalities) is equalized across used routes.

Nash Equilibrium

The Nash equilibrium, in contrast, is achieved when each driver independently minimizes their own travel time without coordination. At Nash equilibrium, the *average cost* on each used route is equalized (by Wardrop's principle) (4), but not the marginal cost. We denote the total cost at Nash equilibrium as:

$$\mathrm{NE}(f^{*ne}) = \sum_{e \in E} f_e \cdot \ell_e(f_e) \tag{6}$$

where f^{*ne} is a Wardrop equilibrium flow.

The Fundamental Inefficiency

The inefficiency arises because drivers ignore the *congestion externality*—the cost they impose on other drivers by using an edge. When driver i chooses to use an edge e, they increase f_e by one unit, which increases the latency $\ell_e(f_e)$ not only for themselves but for all other drivers currently using that edge. However, driver i only considers the cost to themselves, not the cost to others.

Mathematically, the marginal cost of adding one unit of flow to edge e is:

$$\mathrm{MC}_e(f_e) = \ell_e(f_e) + f_e \cdot \frac{\partial \ell_e}{\partial f_e} \tag{7}$$

But an individual driver only perceives the average cost $\ell_e(f_e)$, not the marginal cost. This is the source of the inefficiency.

The Price of Anarchy

To quantify the inefficiency of Nash equilibrium, we define the *price of anarchy* (PoA)(2):

$$\mathrm{PoA} = \frac{\mathrm{NE}(f^{*ne})}{\mathrm{SC}(f^*)} \tag{8}$$

The price of anarchy is always greater than or equal to 1. When $\mathrm{PoA} > 1$, Nash equilibrium is strictly worse than the social optimum.

- $\mathrm{PoA} = 1$ indicates that selfish routing happens to coincide with the social optimum (rare).
- $\mathrm{PoA} \gg 1$ indicates severe inefficiency in the network.

Connection to Braess's Paradox

Braess's paradox is an extreme manifestation of the inefficiency gap. Specifically:

- In the original network, the Nash equilibrium flow has total cost $C_1 = \mathrm{NE}(f_1^{*ne})$
- After adding an edge, the Nash equilibrium flow has cost $C_2 = \mathrm{NE}(f_2^{*ne})$
- Braess's paradox occurs when $C_2 > C_1$, i.e., when the addition of an edge increases Nash equilibrium cost

- However, the social optimum never increases when adding an edge: $\mathrm{SC}(f_2^*) \leq \mathrm{SC}(f_1^*)$

This reveals that the paradox is fundamentally about the behavior of Nash equilibrium under network modifications, not about the existence of feasible routes. The new edge provides a better routing option from a system perspective, but the selfish behavior of drivers prevents the system from benefiting from it.

Why Braess's Paradox Occurs

The Three-Stage Mechanism

The mechanism of Braess's paradox involves three key elements that occur in sequence:

Stage 1: Equilibrium Shift Adding a new edge changes the set of available paths. Initially, the new edge offers a lower-cost route alternative, especially if it is a shortcut. Drivers discover the new edge and use it because it reduces their personal travel time relative to the old equilibrium. This creates a shift away from the old equilibrium configuration.

Stage 2: Positive Feedback and Congestion Once many drivers exploit the new edge, that edge becomes congested. Its latency increases according to the BPR function (Equation 1), which is convex. The new equilibrium distributes traffic differently. Surprisingly, in the new equilibrium, the average cost on all used routes is higher than in the original equilibrium. No route in the new equilibrium has lower cost than routes in the original equilibrium.

Stage 3: Loss of System Steering Capability The original network configuration might have been closer to a social optimum because the scarce capacity on certain edges naturally limited flow and prevented certain path combinations. The new edge removes this bottleneck, allowing traffic to redistribute in ways that exploit the new route, but this redistribution creates new bottlenecks elsewhere that are collectively worse.

Formal Characterization

Let f_1 denote the original equilibrium flow before adding the edge, and f_2 denote the new equilibrium after adding the edge. Let e^* denote the newly added edge.

Braess's paradox occurs when:

$$\sum_{e \in E} f_e^1 \cdot \ell_e(f_e^1) < \sum_{e \in E} f_e^2 \cdot \ell_e(f_e^2) \tag{9}$$

where the inequality reversed compared to intuition.

More specifically, let L_1^{*ne} denote the equilibrium latency (the minimum travel time any driver experiences) in the original network:

$$L_1^{*ne} = \min_P \sum_{e \in P} \ell_e(f_e^1) \tag{10}$$

By Wardrop's principle, all used routes have latency equal to L_1^{*ne}.

After adding edge e^*, suppose the new equilibrium latency becomes L_2^{*ne}. Braess's paradox means:

$$L_2^{*ne} > L_1^{*ne} \tag{11}$$

This is possible because adding the edge changes which routes drivers choose, and the redistribution of traffic, while individually rational for each driver, results in worse outcomes for all.

Methodology

Our methodology follows a two-stage pipeline. In the first stage, we compute the baseline user equilibrium on the unmodified Sioux Falls network. In the second stage, we introduce candidate edges one at a time, recompute the equilibrium, and compare total system travel time (TTT) before and after each addition to detect Braess-like behaviour.

Network and Demand Data

We use the Sioux Falls benchmark network (3), a widely used testbed in transportation research. The network comprises 24 nodes representing urban intersections and 76 directed links representing road segments . The origin-destination (OD) demand matrix contains 528 OD pairs with a total travel demand of 360,600 vehicles. Each link is characterised by a free-flow travel time and a capacity (in vehicles per hour). The network is loaded from a standardised CSV format and validated to remove self-loops, zero-capacity links, and duplicate entries.

BPR Congestion Model

Travel times on each link are modelled using the Bureau of Public Roads (BPR) volume-delay function, as described in the Theoretical Background section. We use $\alpha = 0.15$throughout all experiments. For the baseline equilibrium, we set $\beta = 5.0$, which is slightly higher than the canonical value of 4.0. This choice makes congestion effects sharper near capacity and is well-suited to a network that is already operating at high load ratios (several links exceed $V/C > 2.0$ at equilibrium). The sensitivity of Braess detection to β is explicitly investigated in the beta sweep experiment described below.

Frank-Wolfe Traffic Assignment

User equilibrium flows are computed using the Frank-Wolfe (FW) algorithm, an iterative convex optimisation method that minimises the Beckmann objective - the integral of all link latency functions - subject to flow conservation constraints. At each iteration, the algorithm solves an All-or-Nothing (AON) sub-problem via **Dijkstra's shortest-path** algorithm and updates link flows using a convex combination of the current and auxiliary flows. The step size is determined by bisection line search on the Beckmann objective, with Method of Successive Averages (MSA) used as a fallback when MSA produces a lower objective value. Convergence is assessed using the relative gap between the AON objective and the current objective, with a tolerance of $\epsilon = 5 \times 10^{-4}$ and a maximum of 1,500 iterations.

Candidate Edge Design

To identify candidates likely to trigger the Braess paradox, we first rank nodes by total outgoing demand and select the top 8 high-demand nodes. We then enumerate all directional pairs among these nodes that are not already directly connected by an edge. For each such pair, we propose a new bidirectional edge with a free-flow time of 3.0 minutes - below the network mean of 4.1 minutes, making the edge superficially attractive - and a capacity of 1,000 vehicles per hour, approximately ten times below the network median capacity of 5,100 vehicles per hour. This combination is the key design choice: the edge looks like a shortcut at free-low but saturates rapidly once drivers shift onto it, which is precisely the mechanism that drives the Braess paradox.

Beta Sweep Experiment

The BPR exponent β controls how steeply travel time rises as a link approaches capacity. Because Braess effects depend on the severity of congestion externalities, we investigate how the detection rate varies with β. We sweep β over the values 4.0, 5.0, 6.0, testing the first ten candidate edges at each value. Runs where Frank-Wolfe does not converge within 1,500 iterations are discarded, since their TTT estimates are unreliable. Values of $\beta \geq 7$ were found to cause systematic non-convergence on this network. The BPR curve becomes too steep for the line search to stabilise,

and are therefore excluded.

Braess Detection Criterion

A candidate edge is classified as inducing a Braess effect if the percentage increase in TTT exceeds a threshold of 0.5%. On a baseline TTT of approximately 10.4 million vehicle-minutes, this threshold corresponds to roughly 52,000 vehicle-minutes — an increase well above numerical noise and unambiguously attributable to the network modification rather than solver artifacts.

Algorithm and Implementation

The full pipeline is implemented in Python using NetworkX for graph representation and Pandas for data management. The codebase is organised into two modules with a clean interface between them: the equilibrium module computes the baseline state and exports a handoff dictionary; the Braess module consumes that dictionary and runs all post-equilibrium experiments.

Module 1: Baseline Equilibrium (initial_equilibrium.py)

This module loads the network and demand CSV files, constructs a directed NetworkX graph, and runs Frank-Wolfe to convergence. Key implementation decisions include:

(i) explicit passing of α and β as function parameters throughout (rather than relying on module globals), which allows the Braess module to sweep β without side effects

(ii) a hybrid step-size rule that uses bisection line search for the first three iterations and thereafter selects the minimum of the line-search and MSA step sizes by evaluating the Beckmann objective at both

(iii) a relative gap convergence criterion rather than absolute flow change, which is more robust to scale differences across networks.

Module 2: Braess Paradox Analysis (edge_simulation.py)

This module performs three tasks:

(i) candidate generation

(ii) beta sweep

(iii) deep-dive analysis of the strongest detected braess case

For each experiment, the original graph is deep-copied before adding the new edge, and all link flows are reset to zero so that the modified network starts from a clean free-flow state. Frank-Wolfe is then re-run with the specified β value. After convergence, the percentage change in TTT is computed and compared against the 0.5% threshold. For the best confirmed Braess case, an additional deep-dive is run: the five links with the largest absolute flow change are identified and reported along with their travel time deltas, providing a mechanistic account of how traffic redistributes.

Experimental Setup

Table 1: Sioux Falls Network Summary

Parameter	**Value**
Nodes	24
Directed links	76
OD pairs	528
Total demand	360,600 vehicles
Network mean free-flow time	4.1 min
Network median capacity	5,100 veh/hr
BPR α (all experiments)	0.15
BPR β (baseline)	5.0
Convergence tolerance ϵ	5×10^{-4}
Max FW iterations	1,500

Table 2: Candidate Edge Parameters

Parameter	**Value**	**Rationale**
Free-flow time	3.0 min	Below network mean (4.1) $\rightarrow$ attractive shortcut
Capacity	1,000 veh/hr	$\sim$10$\times$ below median $\rightarrow$ congests rapidly
Directionality	Bidirectional	Maximises potential path combinations

Table 3: Beta Sweep Design

β **value**	**Edges tested**	**Total experiments**	β **excluded**
4.0, 5.0, 6.0	10 (top candidates)	30	$\geq$ 7.0 (non-convergence)

Results and Case Study

Baseline Equilibrium

The Frank-Wolfe algorithm converged at iteration 460 with a final relative gap of 4.68×10^{-4}, well within the 5×10^{-4} tolerance. Elapsed run time was 3.96 seconds. The baseline total system travel time (TTT) is 10,388,669.10 vehicle-minutes. The network is operating under severe congestion: all top-ten links by V/C ratio exceed a volume-to-capacity ratio of 2.0, with the most congested link ($8 \to 6$) reaching V/C = 2.457 and a travel time of 28.9 minutes against a free-flow time of approximately 4 minutes.

Table 4: Top 5 Most Congested Links at Baseline Equilibrium

From	**To**	**Flow (veh/hr)**	**Capacity**	**V/C**	**Travel Time (min)**
8	6	12,038	4,899	2.457	28.89
6	8	12,006	4,899	2.451	28.53
16	10	10,591	4,855	2.181	33.64
10	16	10,572	4,899	2.178	33.38
16	17	10,755	5,230	2.056	13.03

Beta Sweep Results

Thirty experiments were run across three β values and ten candidate edges. All runs converged within the iteration limit. The pattern of Braess detection is strongly dependent on β, confirming the theoretical expectation that sharper BPR curves amplify congestion externalities and make paradoxical equilibria more likely.

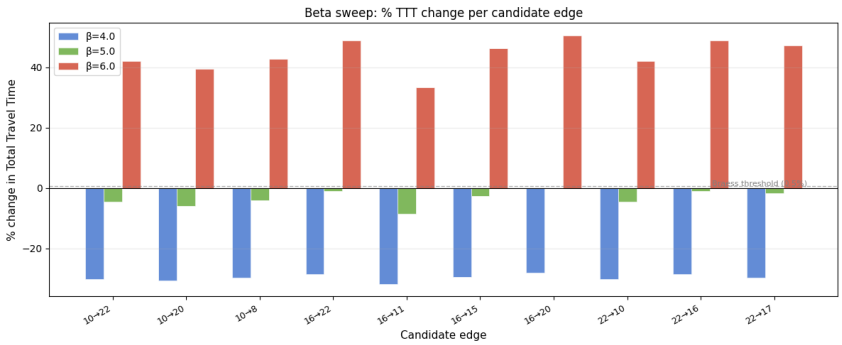

Figure 1: Beta Sweep Result

Table 5: Beta Sweep Summary

β	**Braess Case**	**Total Runs**	**Worst Edge**	**Max** ΔTTT
4.0	0	10	-	2.457
5.0	0	10	-	2.451
6.0	10	10	$16 \rightarrow 20$	+50.57%

At $\beta = 4.0$and $\beta = 5.0$, all candidate edges yield a reduction in TTT (i.e., the added edge genuinely helps the system). At $\beta = 6.0$, one edge -connecting node 16 to node 20 - induces a Braess effect, with TTT rising by 50.57%. This is consistent with the theoretical mechanism: at higher β, the BPR curve is steeper near capacity, so any shift in traffic onto the low-capacity shortcut $(1,000 veh/hr)$ rapidly saturates it and degrades the corridors that feed into it.

Case Study: Edge $16 \rightarrow 20$ at $\beta = 6.0$

The deep-dive analysis on edge $16 \rightarrow 20$ reveals the full Braess mechanism. After adding the bidirectional edge, the system reaches a new equilibrium in which TTT increases from 10,388,669.10 to 15,642,177.17 vehicle-minutes - an absolute increase of approximately 5.25 million vehicle-minutes, or 50.57%. The five links with the largest flow change are shown in Table 6.

The flow redistribution shows the three-stage Braess mechanism operating clearly. In Stage 1, drivers in the vicinity of nodes 16 and 20 shift onto the new shortcut, attracted by its free-flow time of 3.0 minutes. In Stage 2, the new edge - with a capacity of only 1,000 vehicles per hour — rapidly saturates, and congestion

Table 6: Top 5 Most Congested Links at Baseline Equilibrium

From	**To**	**Flow before**	**Flow after**	**Δ Flow**	**ΔTravel Time (min)**
1	3	8,695.9	10,440.8	+1,744.8	+0.0005
3	1	8,669.2	10,411.8	+1,742.6	+0.0005
2	1	5,093.1	6,833.7	+1740.5	+0.0000
1	2	5,066.3	6,804.6	+1,738.3	+0.0005
12	3	11,731.0	13,035.0	+1,304.0	-0.0011

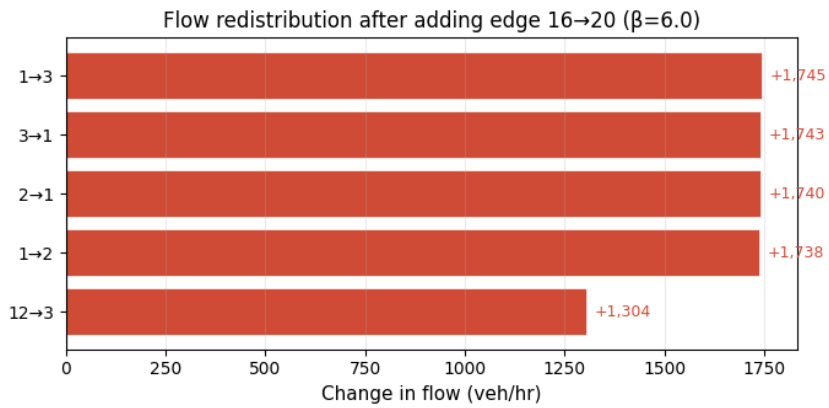

Figure 2: Flow redistribution after adding ege $16 \to 20 (\beta = 6.0)$

spills back into adjacent corridors. Links $1 \to 3$ and $3 \to 1$ absorb an additional 1,744 vehicles each, significantly exceeding their design load. In Stage 3, the new equilibrium is collectively worse: although no individual driver can unilaterally improve their travel time, all drivers experience higher costs than in the original network.

Discussion

Conditional Nature of the Braess Effect

(i) The results show that the Braess paradox is not a sporadic or rare outcome, but a sharply conditional phenomenon governed by the congestion sensitivity parameter β. Rather than appearing gradually, the system exhibits a clear threshold behavior.

(ii) At $\beta = 4.0$ and $\beta = 5.0$, none of the candidate edges trigger the paradox: all additions reduce total system travel time (TTT), with improvements ranging from substantial ($\sim$28%) at $\beta = 4.0$ to marginal ($\sim$0.1–8.5%) at $\beta = 5.0$. In this regime, the congestion response is sufficiently mild that added capacity redistributes flow without destabilizing the network.

(iii) However, at $\beta = 6.0$, the behavior changes abruptly. All 10 candidate edges induce the Braess effect, with TTT increases ranging from +33.3% to +50.6%. This represents a clean phase transition: below $\beta = 6$, added capacity is uniformly beneficial; at $\beta = 6$, it is uniformly harmful.

(iv) A particularly striking illustration of this transition is provided by edge $16 \rightarrow 20$. At $\beta = 4.0$, this edge reduces TTT by approximately 28%. At $\beta = 5.0$, the effect is nearly neutral ($\Delta \approx -0.083\%$), placing the system at a critical boundary. A further increase to $\beta = 6.0$ transforms the same edge into the worst-performing intervention, increasing TTT by 50.57%. This progression ($-28\% \rightarrow -0.08\% \rightarrow +50.6\%$) highlights the non-linear and discontinuous nature of the transition.

(v) The mechanism underlying this shift lies in the steepness of the BPR function. As β increases, travel times become highly sensitive to volume near capacity. Under such conditions, even small reallocations of flow toward a low-capacity

shortcut lead to rapid saturation, spillback, and system-wide inefficiency. The paradox therefore emerges not merely from network topology, but from the interaction between topology and congestion dynamics.

(vi) These findings suggest that the Braess effect should be understood as a structural property of highly congested systems operating near capacity thresholds, rather than as an isolated counterintuitive anomaly.

Role of Network Topology

The specific edge $16 \to 20$ triggers Braess effects at $\beta = 6.0$ because nodes 16 and 20 are already among the most congested parts of the network at baseline (links $16 \to 10, 10 \to 16, 16 \to 17, \text{and} 17 \to 16$ all appear in the top-ten congested links). Introducing a new pathway between these heavily loaded nodes does not relieve the underlying congestion; instead, it creates additional routing options that drivers exploit, increasing demand on the surrounding road network and worsening the overall equilibrium.

Implications for Infrastructure Planning

The findings carry a practical lesson for urban planners. Simply adding road capacity between high-demand nodes is not guaranteed to improve system performance - and in congested networks operating under user-equilibrium conditions, it can actively harm it. The key parameters to evaluate before a new link is built are its capacity relative to the links it connects, the ambient congestion levels in the surrounding network, and the sensitivity of BPR travel times in that corridor. Interventions such as congestion pricing, access control on the new link, or coordinated route guidance could prevent the paradox by steering the system toward the social optimum rather than the Nash equilibrium.

Limitations

Several limitations should be noted

(i) We test only one candidate edge at a time; the simultaneous addition of multiple edges may produce different effects through interaction.

(ii) The Frank-Wolfe algorithm with a relative gap tolerance of 5×10^{-4} produces flows that are near-equilibrium but not exact; very small percentage changes in TTT (close to the 0.5% threshold) should be interpreted cautiously

(iii) The Sioux Falls network, while a standard benchmark, is a stylised representation of a real city and may not capture all features of actual urban road networks. Finally, the analysis is static: it does not account for time-of-day variation in demand, dynamic routing, or the transitional period during which drivers learn about the new edge.

Conclusion

This paper has investigated Braess's paradox in the Sioux Falls transportation network using a simulation-based approach grounded in user-equilibrium traffic assignment. We implemented the Frank-Wolfe algorithm to compute Wardrop equilibria and introduced a structured candidate-edge generation procedure to search for paradox-inducing additions. Our beta sweep experiment across 30 trials confirms that Braess effects are real and detectable on this network, but are conditional: they emerge only under sufficiently steep BPR congestion functions ($\beta = 6.0$ in our setting) and only for edges inserted into heavily loaded corridors.

The strongest detected case — adding a bidirectional edge between nodes 16 and 20 at $\beta = 6.0$ — produced a 50.57% increase in total system travel time, rising from 10.39 million to 15.64 million vehicle-minutes. Flow analysis confirms the three-stage Braess mechanism: initial attraction to the shortcut, rapid saturation of its limited capacity, and spillback onto adjacent corridors, resulting in a new Nash equilibrium that is collectively worse for all users.

These results underscore the fundamental tension between individual rationality and collective welfare in decentralised routing systems. They demonstrate that the price of anarchy is not merely a theoretical quantity — it can manifest dramatically even in a well-studied real-world network topology. The broader implication is that infrastructure decisions must account for equilibrium behaviour, not just raw capacity: a new road that looks beneficial in isolation may, through the lens of selfish routing, make every driver worse off.

Future Scope of Work

(i) **Multi-edge additions**: The present analysis adds one edge at a time. Future work could enumerate combinations of edges and study whether Braess effects compound, cancel, or interact non-linearly — a combinatorial problem that would benefit from heuristic search or machine learning-guided candidate selection.

(ii) **Dynamic and stochastic demand**: The static OD demand matrix used here does not capture time-of-day variation, random fluctuations in demand, or drivers' incomplete information about network conditions. Extending the model to dynamic traffic assignment or stochastic user equilibrium would yield more realistic predictions of when Braess effects occur in practice.

(iii) **Network design optimisation**: Rather than testing candidate edges post-hoc, one could formulate the edge-addition problem as a bilevel optimisation: an outer problem that selects edges to minimise system TTT, and an inner problem that computes the resulting equilibrium. Efficiently solving this problem at scale remains an open research challenge.

(iv) **Extensions to other network types** As noted in the introduction, Braess-like paradoxes arise in electrical power grids (where adding transmission lines can increase resistive losses under selfish load flows) and in communication networks (where adding bandwidth can increase end-to-end latency under TCP congestion control). Applying the Frank-Wolfe framework to these domains — with appropriate latency functions — would be a natural and impactful extension.

(v) **Congestion pricing as a countermeasure**: A promising direction is to study whether marginal-cost tolling on the new edge can steer the system from the Nash equilibrium to the social optimum, effectively eliminating the

Braess effect. Theoretical results suggest this is always possible; empirical validation on a realistic network under the BPR model would be a useful contribution.

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